Eventive Carpooling – Final Year Project Summary

Project Title: Eventive Carpooling – A Route Propagation Based Dynamic Carpooling System

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Abstract

Eventive Carpooling is a smart mobility solution designed to address urban traffic congestion, transportation costs, and environmental concerns through a flexible, route-aware carpooling system. Unlike conventional carpooling platforms that match riders with identical destinations, our system implements dynamic route propagation—allowing drivers to accept new passengers along their ongoing route, essentially enabling a ride that never ends.

This intelligent system supports multi-hop trips where a passenger can reach a destination using a chain of rides, based on real-time availability. Users can plan, book, and pay for rides through a mobile application built with React Native, powered by Node.js, Firebase, and machine learning-based route matching. Safety features such as gender-specific ride options and live tracking are included to build user trust and broaden adoption.

Key Features

- Dynamic route propagation — continuous matching of riders with drivers along the route.
- Multi-hop ride chaining — passengers can complete journeys using a sequence of compatible rides.
- Live location tracking via Google Maps API.
- Gender-specific carpooling for enhanced safety.
- Ride cost sharing, route optimization, and environmental benefits.
- Built-in event creation: drivers can schedule rides with specific time, destination, and seat availability.

Technology Stack

- Frontend: React Native (for Android)
- Backend: Node.js, Firebase (Authentication & Realtime DB)
- Database: MongoDB

- Machine Learning: Route similarity and fare prediction
- APIs: Google Maps API (route mapping), Online Payment Integration

Problem Statement

Most current carpooling solutions only connect drivers and riders with exactly matching origins and destinations. This significantly limits the potential of ride-sharing. Moreover, female users often feel insecure using mixed-gender ride-sharing services. Our platform solves both challenges:

- By enabling non-identical route-based matching using intelligent algorithms.
- By offering gender-aware pooling options to make travel safer and more inclusive.

Implementation Highlights

- Route Matching Algorithm: Combines Google Maps routing with Dijkstra's algorithm to identify optimal ride sequences.
- Heuristic and Euclidean Distance Methods: For fast and efficient location matching.
- Ride Event Casting: Drivers can create ride events, the system evaluates passenger interest in real-time.
- Safety First: Firebase authentication, ride tracking, and gender-filtered ride options are incorporated.

System Architecture Overview

- Mobile App: Interface for passengers and drivers.
- Admin Portal (Web): Admin-level monitoring and management.
- Core Engine: Logic for ride propagation, chaining, and ML-based suggestions.

Benefits

- Cuts down fuel usage and cost by enabling shared rides.
- Reduces city congestion by decreasing single-occupancy vehicle usage.
- Promotes a cleaner environment through fewer emissions.
- Encourages community engagement by connecting users.
- Provides a safe, affordable, and scalable transport solution.

Evaluation and Testing

The system was tested for:

- Functionality: Real-time ride creation, matching, and payment workflow.
- Usability: Smooth UI/UX and minimal onboarding steps.
- Compatibility: Working on major Android versions and across devices.
- Performance: Stress-tested under high user load.
- Security: Data encryption and secure authentication.
- Feedback Loop: User feedback used for continual improvements.

Conclusion

Eventive Carpooling introduces an innovative shift in ride-sharing paradigms by integrating continuous ride propagation, safety-aware matching, and dynamic routing intelligence. It redefines urban commuting with a focus on sustainability, accessibility, and user-centric design. The project demonstrates technical feasibility and real-world applicability, especially in urban areas facing transit bottlenecks.